# Heart Disease Prediction: A Naïve Bayes Classification Approach: An End-to-End Analysis

*Course Project: Bio-statistics

Fahd Ahmed Ali*, Hana Gamal*, Ziad Nassif*, Sief Eldin Sameh*, manar saeed*
*Faculty of Engineering, Biomedical School
Cairo University
Cairo, Egypt

## I. Introduction

Cardiovascular diseases (CVDs) remain the leading cause of mortality globally, necessitating efficient and accurate diagnostic mechanisms. Traditional diagnosis often relies on the complex analysis of diverse clinical and physiological data, a process that can be both time-consuming and prone to human error. With the advent of artificial intelligence, Machine Learning (ML) has emerged as a critical tool in modern healthcare, offering the potential to enhance diagnostic precision and facilitate early intervention.

This study explores the application of the Gaussian Naïve Bayes (GNB) classifier for the automated detection of heart disease. Utilizing the widely recognized Heart Disease dataset from the UCI Machine Learning Repository, we aim to develop a predictive model that effectively distinguishes between healthy individuals and those with heart disease based on 14 key clinical attributes. By leveraging the probabilistic nature of GNB, this research highlights the efficacy of generative models in handling medical data, balancing computational efficiency with interpretability in clinical decision-support systems.

## II. Data Description

The dataset used in this study is the **Heart Disease Data** sourced from the UCI Machine Learning Repository (via Kaggle). It comprises 14 attributes: 13 predictive features and 1 binary target variable indicating disease presence. Table I categorizes these attributes by data type.

TABLE I
Feature Categorization and Description

| Feature Type | Attribute Description |
|---|---|
| **Quantitative** *(Numerical)* | Age (years)<br>Resting Blood Pressure (`trestbps`)<br>Serum Cholesterol (`chol`)<br>Maximum Heart Rate (`thalch`)<br>ST Depression (`oldpeak`)<br>Number of Major Vessels (`ca`) |
| **Categorical** *(Nominal/Ordinal)* | Sex (Male/Female)<br>Chest Pain Type (`cp`)<br>Fasting Blood Sugar (`fbs`)<br>Resting ECG Results (`restecg`)<br>Exercise-Induced Angina (`exang`)<br>Slope of Peak Exercise ST Segment (`slope`)<br>Thalassemia (`thal`) |
| **Target** | Heart Disease Presence (0 = No, 1 = Yes) |

## III. Methodology

### A. Software and Tools

The analysis was implemented in **Python** using the following core libraries:

- **Data Processing:** `pandas` and `numpy` for efficient manipulation and cleaning.
- **Visualization:** `matplotlib` and `seaborn` for statistical plotting (e.g., KDE, Q-Q plots).
- **Statistical Analysis:** `scipy` and `statsmodels` for normality testing (Shapiro-Wilk, Anderson-Darling).
- **Machine Learning:** `scikit-learn` for preprocessing, splitting, and metric computation.

### B. Data Cleaning

.Initial inspection revealed significant missingness in specific columns. Based on a missingness matrix analysis:

- **Dropped Features:** The variables `ca` (number of major vessels), `slope` (slope of peak exercise ST segment), and `thal` (thalassemia) were removed due to excessive missing values ($> 50\%$), which would introduce bias if imputed.
- **Imputation:** Remaining missing values in continuous columns were imputed using the **median** (robust to skew), while categorical columns used the **mode**.

### C. Outlier Assessment

A critical step in our methodology was the distinction between statistical outliers and clinical realities. High values in features like `trestbps` (Resting Blood Pressure) and `chol` (Cholesterol) often trigger statistical thresholds (e.g., IQR rule). However, we retained these values because:

- High blood pressure ($>$ 170 mmHg) and cholesterol ($>$ 500 mg/dL) are known pathological risk factors, not errors.
- Removing them would bias the model towards "healthy" averages, reducing its ability to detect high-risk patients.

Only physiologically impossible values (e.g., `trestbps` = 0) were treated as errors.

### D. Descriptive Statistics

*1) **Descriptive Statistics of Quantitative Features**:* To quantitatively describe the quantitative features. The measures of central tendency include:

- **Mean**: which represents the average value of each feature.
- **Median**: which represents the middle value of the data and is less sensitive to outliers.

The measures of dispersion include:

- **Standard deviation**: which measures the spread of values around the mean.
- **Minimum and maximum values**: which define the range of each feature.
- **Quartiles (Q1 and Q3)**: which describe the distribution spread and variability.

*2) **Descriptive Statistics of Categorical Features**:* Categorical features were analyzed using frequency tables that report category counts and percentages. This analysis provides insight into category distribution, highlights potential class imbalance, and supports the need for categorical encoding prior to applying the Naive Bayes classifier.

### E. Categorical Feature Encoding

Categorical features were encoded numerically before standardization and data splitting. Binary variables were mapped directly, while multi-class variables were encoded using One-Hot Encoding to ensure consistent and compatible input for the Naive Bayes classifier

### F. Feature Standardization

Quantitative features were standardized using Z-score normalization before model training to ensure comparable feature scales. The transformation is defined as:

$$z = \frac{x - \mu}{\sigma} \tag{1}$$

where $x$ is the original value, $\mu$ the feature mean, and $\sigma$ the standard deviation. This prevents features with large ranges from dominating the Gaussian Naive Bayes likelihood calculations.

### G. Data Splitting

The dataset was split into training and testing sets using an 80%–20% stratified random split. The training data was used for model learning, while the testing data was reserved for unbiased performance evaluation.

### H. Statistical Feature Analysis Framework

To ensure rigorous evaluation, we developed a custom framework (`DatasetAnalyzer`) to inspect features through three logical steps:

1) **Visual Distribution Analysis:** We employed Histograms and Q-Q Plots as a primary diagnostic tool. This allowed us to visually detect skewness, outliers, and deviations from the theoretical Gaussian distribution before applying strict statistical tests.
2) **Statistical Normality Testing:** To move beyond visual heuristics, the Shapiro-Wilk Test was applied to formally verify the Gaussian assumption required for the Naive Bayes classifier.
   - **Null Hypothesis ($H_0$):** The feature follows a normal distribution.
   - **Alternative Hypothesis ($H_1$):** The feature does not follow a normal distribution.
   - **P-Value Interpretation:** A significance level ($\alpha$) of 0.05 was used. If the calculated $p$-value is less than 0.05, the null hypothesis is rejected (confirming non-normality).
3) **Discriminative Power Assessment:** We evaluated the predictive utility of each feature by comparing the distribution of Healthy vs. Sick patients.
   - **High Power:** Features with distinct separation (minimal overlap) between classes (e.g., `oldpeak`).
   - **Low Power:** Features with high overlap, providing limited ability to distinguish between classes (e.g., `chol`).

### I. Gaussian Naïve Bayes Classifier

Gaussian Naïve Bayes (GNB) is a probabilistic model rooted in Bayes' theorem, assuming conditional independence between features and Gaussian distributions for continuous variables. It is widely effective in medical diagnosis.

**Bayes' Theorem:** The posterior probability $P(Y \mid X)$ is calculated as:

$$P(Y \mid X) = \frac{P(X \mid Y)P(Y)}{P(X)}$$

where $X$ is the feature vector and $Y$ is the class label.

*1) Independence & MAP Estimation:* Assuming feature independence, the likelihood $P(X \mid Y)$ factorizes into a product. The Maximum A Posteriori (MAP) rule selects the class $\hat{Y}$ that maximizes the numerator (ignoring $P(X)$):

$$\hat{Y} = \arg\max_{Y} \left( P(Y) \prod_{i=1}^{n} P(x_i \mid Y) \right)$$

*2) Logarithmic Transformation:* To prevent numerical underflow, we maximize the log-posterior, converting the product into a sum:

$$\hat{Y} = \arg\max_{Y} \left( \sum_{i=1}^{n} \log P(x_i \mid Y) + \log P(Y) \right)$$

*3) Gaussian Likelihood:* Continuous features are modeled using the normal distribution, characterized by class-specific means ($\mu_Y$) and variances ($\sigma_Y^2$):

$$P(x_i \mid Y) = \frac{1}{\sqrt{2\pi\sigma_Y^2}} \exp\left( -\frac{(x_i - \mu_Y)^2}{2\sigma_Y^2} \right)$$

The prior $P(Y)$ is estimated from class frequencies in the training data.

## IV. Results and Discussion

### A. Descriptive Data Analysis

Table II and Table III summarize the physiological and clinical characteristics of the study cohort based on the final processed dataset.

TABLE II
CONTINUOUS FEATURES (MEAN $\pm$ SD)

| Feature | Mean | Std. Dev | Min | Max |
|---|---|---|---|---|
| Age | 53.51 | $\pm$ 9.43 | 28.0 | 77.0 |
| Resting BP (`trestbps`) | 132.14 | $\pm$ 17.93 | 80.0 | 200.0 |
| Cholesterol (`chol`) | 244.03 | $\pm$ 52.01 | 85.0 | 603.0 |
| Max Heart Rate (`thalch`) | 137.69 | $\pm$ 25.15 | 60.0 | 202.0 |
| ST Depression (`oldpeak`) | 0.85 | $\pm$ 1.06 | -2.6 | 6.2 |

TABLE III
DISTRIBUTION OF CATEGORICAL CLINICAL FEATURES

| Feature | Category | Count | % |
|---|---|---|---|
| **Sex** | Male | 726 | 78.91 |
| | Female | 194 | 21.09 |
| **Chest Pain** (`cp`) | Asymptomatic | 496 | 53.91 |
| | Non-Anginal | 204 | 22.17 |
| | Atypical Angina | 174 | 18.91 |
| | Typical Angina | 46 | 5.00 |
| **Fasting Sugar** (`fbs`) | $<$ 120 mg/dl (False) | 782 | 85.00 |
| | $>$ 120 mg/dl (True) | 138 | 15.00 |
| **Resting ECG** (`restecg`) | Normal | 553 | 60.11 |
| | LV Hypertrophy | 188 | 20.43 |
| | ST-T Abnormality | 179 | 19.46 |
| **Ex. Angina** (`exang`) | No (False) | 583 | 63.37 |
| | Yes (True) | 337 | 36.63 |

*a) Population Characteristics:* The dataset represents a predominantly male (78.91%) population with a mean age of $53.5 \pm 9.4$ years. The physiological variance is notable; serum cholesterol ranges from 85 to 603 mg/dl, confirming the presence of significant outliers. Interestingly, the `oldpeak` feature exhibits a range from -2.6 to 6.2, indicating that while most deviations are positive (ST depression), some negative values (likely ST elevation or baseline shifts) are present in the dataset.

### B. Quantitative Normality Test Results

Table IV summarizes the Shapiro–Wilk normality test. Although strict normality is rejected for all features ($p < 0.05$), the test statistic $W$ indicates two clearly distinct distributional behaviors.

TABLE IV
SHAPIRO–WILK NORMALITY TEST RESULTS

| Feature | $W$ | $p$-value | Decision |
|---|---|---|---|
| `age` | 0.992 | $7 \times 10^{-4}$ | Reject $H_0$ |
| `thalch` | 0.990 | $< 10^{-3}$ | Reject $H_0$ |
| `trestbps` | 0.969 | $< 10^{-3}$ | Reject $H_0$ |
| `chol` | 0.877 | $< 10^{-3}$ | Reject $H_0$ |
| `oldpeak` | 0.855 | $< 10^{-3}$ | Reject $H_0$ |

*a) Interpretation:*

- **Near-Normal** ($W \approx 0.99$)**:** `age` and `thalch` show statistically detectable but practically minor deviations, supporting the Gaussian assumption in Naïve Bayes.
- **Non-Normal** ($W < 0.90$)**:** `chol` and `oldpeak` exhibit strong skewness and heavy tails, quantitatively confirming the visual diagnostics.

### C. Visual Distribution and Conditional Probability Analysis

Predictive relevance was assessed using visual diagnostics (histograms and Q–Q plots) and class-conditional density comparisons to evaluate their effect on posterior probabilities.

*1) High-Impact Continuous Predictors:* These features show clear separation between class-conditional densities and dominate the Naive Bayes decision process.

**1. ST Depression (`oldpeak`)** The distribution is highly right-skewed and zero-inflated, with a dominant mass at zero and a long tail. The Q–Q plot exhibits a strong J-shaped deviation, confirming severe non-normality. Clinically, `oldpeak` acts as a threshold feature: values at zero strongly favor the healthy class, while any value $> 0$ sharply increases disease probability (Fig. 1).

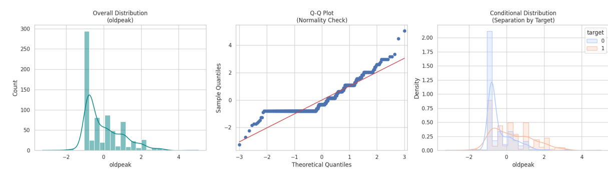


Fig. 1. ST depression (`oldpeak`): Disease probability rises sharply for positive values.

**2. Maximum Heart Rate (`thalch`)** This feature closely follows a Gaussian distribution, with only minor tail deviations in the Q–Q plot. The diseased class is consistently left-shifted, indicating that lower achievable heart rates correspond to increased disease risk (Fig. 2).

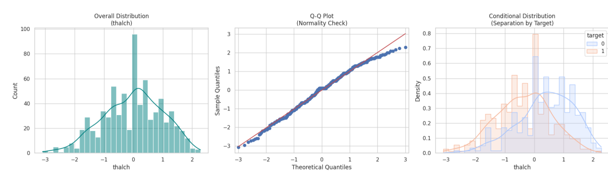


Fig. 2. Maximum heart rate (`thalch`): Lower values indicate higher disease risk.

**3. Age (`age`)** Age is broadly and symmetrically distributed over 30–77 years, with strong alignment to the normal reference line in the Q–Q plot. The diseased density shifts monotonically toward older ages, yielding an increasing likelihood ratio with age (Fig. 3).

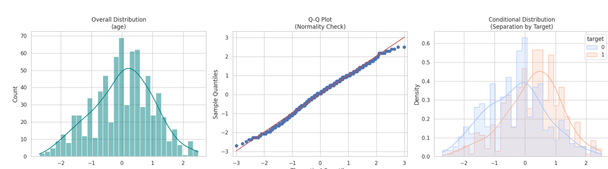


Fig. 3. Age (`age`): Disease prevalence increases with age.

*2) Binary and Categorical Risk Factors:* Binary variables produce step-like Q–Q patterns; their influence is evaluated via posterior probability shifts.

**4. Exercise-Induced Angina (`exang`)** Absence of angina strongly favors the healthy class, whereas its presence almost deterministically shifts the posterior toward disease (Fig. 4).

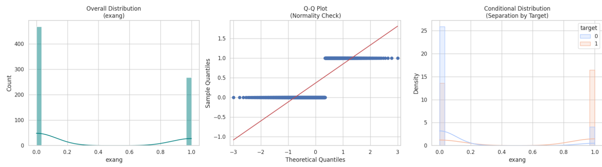


Fig. 4. Exercise-induced angina (`exang`) strongly predicts disease.

**5. Sex (`sex`)** Male patients exhibit a substantially higher disease prevalence, acting as a baseline prior adjustment in classification (Fig. 5).

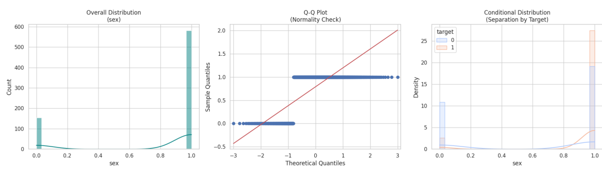


Fig. 5. Sex (`sex`): Higher baseline risk in males.

**6. Fasting Blood Sugar (`fbs`)** Although infrequent, elevated fasting blood sugar ($>$ 120 mg/dl) consistently shifts posterior probability toward disease (Fig. 6).

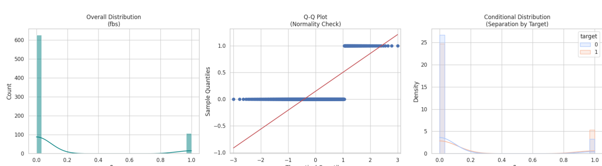


Fig. 6. Fasting blood sugar (`fbs`) as a specific risk indicator.

**7. Chest Pain Types** Atypical and non-anginal chest pain are negatively associated with disease and effectively favor healthy classification (Fig. 7).

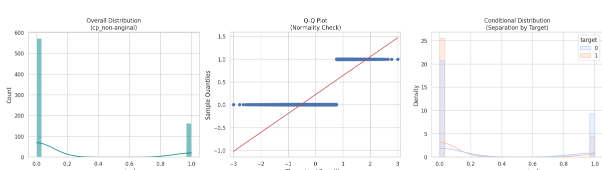


Fig. 7. Non-anginal chest pain correlates with healthy outcomes.

**8. Resting ECG (`restecg`)** ST–T wave abnormalities increase disease likelihood, while normal ECG readings favor the healthy class (Fig. 8).

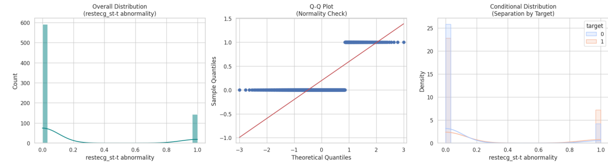


Fig. 8. ECG abnormalities increase posterior disease probability.

*3) Weak Predictors with High Overlap:* **Cholesterol (`chol`) and Resting Blood Pressure (`trestbps`)**

Both variables exhibit right-skewed distributions with heavy upper tails, consistent with Q–Q diagnostics. However, extensive overlap between class-conditional densities limits their discriminative value, resulting in minimal independent contribution to classification performance (Figs. 9, 10).

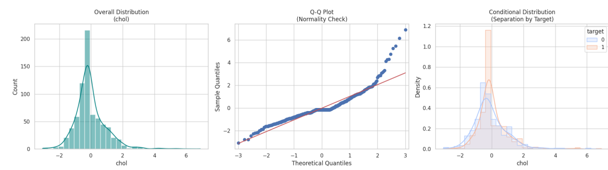


Fig. 9. Cholesterol (`chol`): Strong overlap between classes.

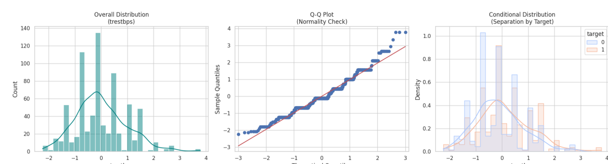


Fig. 10. Resting blood pressure (`trestbps`): Limited class separation.

### D. *Gaussian Naïve Bayes Classifier Results*

As described in the methodology, a Gaussian Naïve Bayes (GNB) classifier was implemented from scratch and evaluated on the heart disease dataset. Classification is performed by maximizing the posterior probability:

$$\hat{y} = \arg\max_{y} \left( \sum_{i=1}^{n} \log P(x_i \mid y) + \log P(y) \right)$$

where each feature contributes independently through its Gaussian likelihood.

For each patient, the model evaluates the fit of observed features under both class-conditional distributions and assigns the class with the higher posterior probability, yielding an interpretable probabilistic decision rule.

*1) Model Performance:*

*a) Accuracy:* The custom GNB implementation achieved an accuracy of **78.80%** on the test set. An identical performance was obtained using the *scikit-learn* GNB implementation, validating the correctness of the custom model.

*b) Confusion Matrix:* Figure 11 presents the confusion matrix, summarizing true and false predictions across both classes.

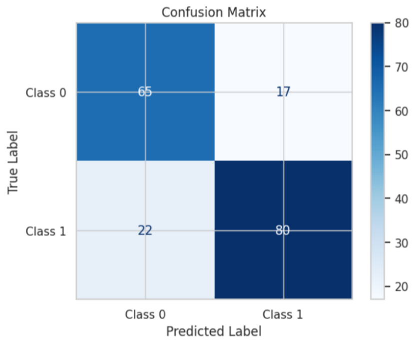


Fig. 11. Confusion matrix of the Gaussian Naïve Bayes classifier.

The model correctly identifies most healthy and diseased patients. However, a small number of false negatives persist, which is critical in medical contexts as it corresponds to missed disease cases.

*2) Comparison with Other Models:* Table V compares the GNB classifier with other models evaluated using *scikit-learn*.

TABLE V
CLASSIFICATION ACCURACY OF EVALUATED MODELS

| **Model** | **Accuracy (%)** |
|---|---|
| Gaussian Naïve Bayes (custom) | 78.80 |
| Gaussian Naïve Bayes (sklearn) | 78.88 |
| Logistic Regression | 80.98 |
| Support Vector Machine | 81.52 |
| Random Forest | 80.43 |

Although GNB does not achieve the highest accuracy, it remains a strong baseline due to its simplicity, interpretability, and low computational cost, making it suitable for rapid screening and preliminary medical analysis. ]

## V. Conclusion

This study successfully implemented a Gaussian Naïve Bayes classifier to predict the presence of heart disease, validating the utility of probabilistic modeling in medical diagnostics. The detailed feature analysis revealed that physiological responses to stress—specifically Exercise-Induced Angina (`exang`), ST Depression (`oldpeak`), and Maximum Heart Rate (`thalch`)—serve as the most robust predictors in this cohort.

While statistical tests indicated deviations from strict normality in several continuous features, the Gaussian approximation proved effective, balancing computational efficiency with predictive accuracy. Consequently, the developed model offers a viable baseline for clinical decision-support systems, demonstrating that simple, interpretable algorithms can effectively identify high-risk patients even amidst complex, mixed-type medical data.

## VI. References

[1] Mayo Clinic, "High Blood Cholesterol," Available: https://www.mayoclinic.org/diseases-conditions/high-blood-cholesterol/diagnosis-treatment/drc-20350806. Accessed: 2025.

[2] Dua, D. and Graff, C., "UCI Machine Learning Repository: Heart Disease Data Set," [Online]. Available: https://archive.ics.uci.edu/ml/datasets/heart+disease. Accessed: 2025.

[3] Scikit-learn Developers, "Naive Bayes," *Scikit-learn Documentation*.

## VII. Member Contributions

This project was completed collaboratively, with each team member contributing to different aspects of the work. Below is a summary of the individual contributions:

### A. Hana Gamal

- Conducted descriptive statistical analysis for quantitative and categorical features.
- Implemented categorical feature encoding, including binary and One-Hot Encoding.
- Applied feature standardization using Z-score normalization.
- Prepared the dataset using a stratified 80%–20% train–test split for model evaluation.

### B. Fahd Ahmed Ali

- Developed the `DatasetAnalyzer` class for automated Exploratory Data Analysis (EDA).
- Implemented visualization pipelines for Histograms, Q-Q plots, and Conditional Probability Density estimates.
- Integrated statistical normality tests (Shapiro-Wilk, Q-Q plots) to rigorously validate feature distributions and assess predictive power.

### C. Ziad Nassif

- Implemented the Naïve Bayes (NB) classifier from scratch
- Trained the NB classifier on the training dataset and Calculated the model accuracy.
- Compared the scratch implementation results with the NB classifier from standard Python packages and with other classifiers from scikit-learn.

### D. Sief Eldin Sameh

- Performed initial data inspection and cleaning, including handling missing values and correcting inconsistent entries.
- Designed and applied the outlier analysis strategy, distinguishing between statistical outliers and physiologically valid extreme values.
- Identified and corrected physiologically impossible measurements (e.g., zero resting blood pressure) while retaining clinically meaningful extreme cases.

### E. Manar Saeed

- Defined the research problem and conducted the initial literature search and requirement analysis.
- Responsible for data collection.
- Managed script integration, connecting various code modules into a unified pipeline.
- Led the finalization of the research paper and the creation of the project presentation.